

Ex02. Introduction to Signals and systems

s1320112/ Motohisa YAMANO

1.1. Develop 7 charts of LTI system function

This assignment involves generating seven charts that illustrate the fundamental behavior of **LTI systems** as described in the problem set. The discrete signals are defined with a variable τ ranging from -10 to 10 with an increment of 0.01. Assuming the constant $a = 0.5$, this section analyzes the behaviors of $x(\tau)$ and $h(\tau)$, and visualizes the resulting convolution $y(t)$.

```
a = 0.5;
tau = -10:0.01:10;
x = @(t) heaviside(t);
h = @(t) exp(-a .* t) .* heaviside(t);

figure('Position', [100, 100, 600, 1000]);

subplot(4,2,1);
p1 = plot(tau, x(tau));
title('1. x(\tau) signal');
xlabel("\tau");
ylabel("Amplitude");
set(p1, 'Color', 'r');
set(p1, 'Marker', '^');
grid on;

subplot(4,2,2);
p2 = plot(tau, h(tau));
title('2. h(\tau) signal');
xlabel("\tau");
ylabel("Amplitude");
set(p2, 'Color', 'r');
set(p2, 'Marker', '^');
grid on;

t = -2
```

```
t =
-2
```

```
subplot(4,2,3);
p3=plot(tau, x(t-tau));
title(['3. x(', num2str(t), '-\tau)']);
xlabel("\tau");
ylabel("Amplitude");
set(p3, 'Color', 'g');
set(p3, 'Marker', '+');
grid on;
```

```

subplot(4,2,4);
p4=plot(tau, h(t-tau));
title(['4. h(', num2str(t), '-\tau)']);
xlabel("\tau");
ylabel("Amplitude");
set(p4, 'Color', 'g');
set(p4, 'Marker', '+');
grid on;

```

```
t = 3
```

```

t =
3

```

```

subplot(4,2,5);
p5=plot(tau, x(t-tau));
title(['5. x(', num2str(t), '-\tau)']);
xlabel("\tau");
ylabel("Amplitude");
set(p5, 'Color', 'b');
set(p5, 'Marker', 'x');
grid on;

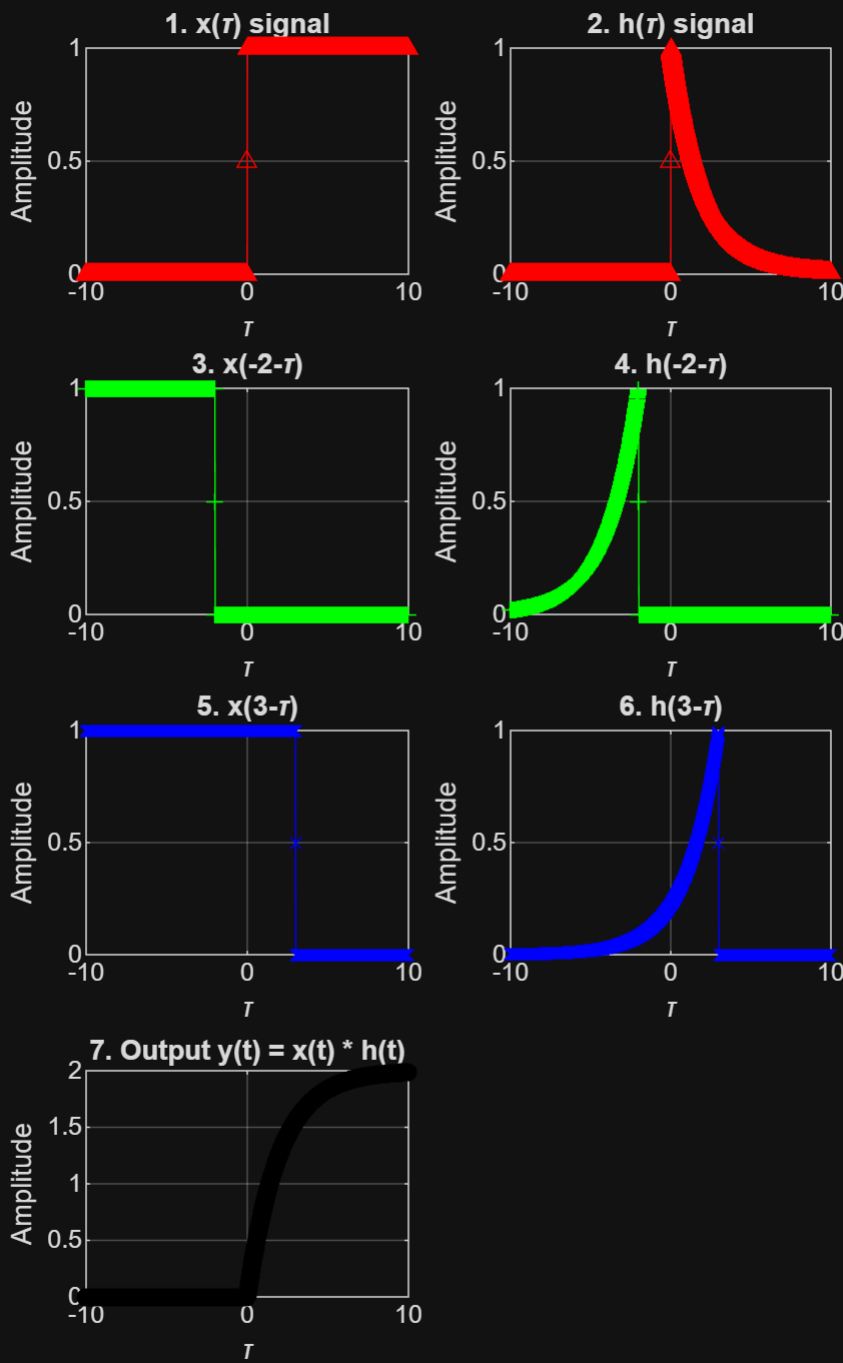
subplot(4,2,6);
p6=plot(tau, h(t-tau));
title(['6. h(', num2str(t), '-\tau)']);
xlabel("\tau");
ylabel("Amplitude");
set(p6, 'Color', 'b');
set(p6, 'Marker', 'x');
grid on;

subplot(4,2,7);
y =(1/a) * (1 - exp(-a .* tau)) .* (tau >= 0);
subplot(4,2,7);
p7 = plot(tau, y);
title('7. Output y(t) = x(t) * h(t)');
xlabel("\tau");
ylabel("Amplitude");
set(p7, 'Color', 'k');
set(p7, 'Marker', 'o');
grid on;

%% Below is the 7 graph required in the assignemnt.
sgtitle('LTI System Analysis');

```

LTI System Analysis



1.2. Math Problem

Below is the snapshot of my answer in fill-blank math problem.

do not overlap //

overlap during the period $[0, t]$ //

0 //

$$\int_0^t e^{-a(t-\tau)} d\tau //$$

$$\frac{1}{a}(1 - e^{-at})$$